

Sea Wave Modelling and Prediction Based on Multisensor Fusion

Final Report of the Integrative Project

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1 Introduction

1.1 Context

- Introduce the purpose of the report.
- Present the organizational context in which the project/internship took place (company, institution, research unit, research laboratory, etc.).
- Present the problem addressed and the motivation for the work carried out.

This internship is carried out within the scope of the SEAWINGS and AIR-SHIP projects, initiatives aimed at developing innovative autonomous aerial platforms specifically designed to operate over maritime environments. The distinguishing feature of these platforms is their design as Wing-In-Ground (WIG) effect vehicles. These vehicles stand out for the operational requirement of flying at high speeds and extremely low altitudes, often only one or two metres above the water surface. This flight configuration takes advantage of the aerodynamic "ground effect" phenomenon, creating a cushion of air beneath the structure that drastically increases the vehicle's lift and energy efficiency.



Figure 1: Example of a vehicle operating under the Wing-In-Ground (WIG) effect principle.

Because of this requirement for fast, low-altitude flight, one of the central and most critical challenges of these projects is accurate, real-time perception of the sea surface. In an environment where the margin for error is minimal and the fluid dynamics are unpredictable, timely detection of waves is not merely a matter of topographic mapping, but a fundamental safety requirement. The ability to "see" the water ahead of the vehicle ensures that the control systems can perform evasive manoeuvres or altitude adjustments, avoiding catastrophic collisions with the swell.

Ocean waves are a dynamic and periodic phenomenon, characterized mainly by their amplitude, direction and frequency. Correctly estimating the geometric and kinematic values of these parameters, as well as the ability to predict the short-term temporal and spatial propagation of waves, contributes significantly to the robustness of the control and trajectory-planning systems of these platforms under development.

1.2 Objectives and expected results

State the objectives of the work and the expected results.

O principal objetivo do estágio é o estudo, desenvolvimento e validação de abordagens de percepção e modelização de ondas oceânicas, baseadas em arquiteturas de fusão multissensoria adaptadas às limitações e requisitos operacionais de plataformas aéreas autónomas de Efeito de Solo (WIG).

1.3 Report structure

Explain the structure of the report.

2 Methodology and main activities carried out

2.1 Methodology used

Describe the methodology[1] followed (e.g. iterative development with biweekly sprints and weekly follow-up meetings) and the resources used (e.g. GitHub [2], etc.).

2.2 Stakeholders, roles and responsibilities

Identify the project team, stakeholders and other parties with whom there was interaction; in the case of group work, clarify the roles and responsibilities of each group member.

2.3 Activities carried out

Describe the activities carried out over time (including relevant events such as presentations, client meetings, etc.) and the respective deliverables, typically using a Gantt chart[3] and a summary description of each activity/deliverable.

3 Solution development

3.1 Requirements

Identify the relevant functional and non-functional requirements and their sources, as well as project constraints.

3.2 Architecture and technologies

Architecture and technologies used and their justification, technical diagrams produced, technical difficulties encountered and their resolution, etc.

3.3 Developed solution

Present the developed solution from the user's perspective, with the help of screenshots.

3.4 Validation

Description of the validation of the developed solution (e.g. experimental evaluation results, tests performed, feedback from users or experts, etc.).

4 Conclusions

4.1 Results achieved

Summarize the results achieved and contributions (with respect to the objectives).

4.2 Lessons learned

Reflect on the lessons learned (taking into account the learning objectives).

4.3 Future work

Ideas for improvements and future work.

4.4 SDG impact

If applicable, identify the contribution of the work to the Sustainable Development Goals (SDGs) established by the United Nations (<https://sdgs.un.org/goals>), with a view to a more sustainable and inclusive future.

References

- [1] Mihai Liviu Despa. Comparative study on software development methodologies. *Database Systems Journal*, 5(3):37–56, 2014.
- [2] GitHub Inc. Github. <https://github.com/>, 2023. [Online; acedido 3 de março de 2023].
- [3] wikipedia. Gantt chart. https://en.wikipedia.org/wiki/Gantt_chart, 2023. [Online; acedido 3 de março de 2023].